
DUAL-AUDIOBENCH: Closed-Loop Evaluation of Belief-State Synchronization in Audio-Conditioned Conversational Agents

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Abstract

1 Reliable voice agents must update their view of the world when their actions,
2 elapsed time, external events, or users change it. Current evaluations measure spo-
3 ken task completion, interruption handling, and stateful tool use, yet rarely reveal
4 whether an agent’s internal belief follows a delayed hidden transition. We intro-
5 duce DUAL-AUDIOBENCH, a turn-based audio-replay benchmark for long hori-
6 zon state synchronization evaluation in voice agents. Each evaluation run places
7 an agent in a deterministic environment, executes its first action, advances simu-
8 lated time, and then resumes with a spoken observation that leaves the new state
9 partly implicit. The benchmark contains 84 scenarios organized into 42 matched
10 causal pairs across 14 domains and three clue distances. Within each pair, one
11 early fact changes the terminal state and correct final action with fixed post-gap
12 words and action menus. Models report probabilities over possible states before
13 and after the gap, allowing belief updating to be measured separately from ac-
14 tion choice. Across 4,368 completed trajectories with three audio-capable models,
15 first-action accuracy under audio replay ranged from 65.5% to 80.4%, but final-
16 action accuracy after the hidden transition ranged from 22.0% to 47.0%. Sup-
17 pressing the transition improved immediate state estimates by 21.4–32.1 points,
18 whereas moving the clue close to the decision did not help. Explicitly stating the
19 changed state improved final-action accuracy by 32.7–41.1 points. These controls
20 pinpoint the main difficulty as inferring an unobserved transition instead of retain-
21 ing the clue alone.

22 1 Introduction

23 Voice agents increasingly make decisions that change external systems. The result of an action
24 may remain unobserved while a maintenance job runs, a transaction settles, or a user independently
25 modifies the same account. When dialogue resumes, the agent must infer the current state before
26 choosing its next action. Final task success alone cannot show whether the agent tracked the transi-
27 tion: the same endpoint can arise from a calibrated decision, a lucky guess, or a stale belief paired
28 with an inconsistent action.

29 Audio-language benchmarks cover acoustic understanding, paralinguistics, long recordings, and
30 multi-turn memory [Yang et al., 2024, Wang et al., 2024, Li et al., 2026, Gosai et al., 2025].
31 VoiceAgentBench adds multilingual tool invocation and adversarial spoken queries [Jain et al.,
32 2025]. Full-duplex evaluations measure timing, overlap, correction, and interruption [Lin et al.,
33 2025b,a, Ge et al., 2025]. EchoChain is the closest state-update benchmark: it tests whether an
34 agent revises an ongoing response after a user interrupts mid-speech [Modi et al., 2026]. Its target is
35 online revision during concurrent speech. Our target is cross-gap inference after an action-dependent

36 world transition, with the agent’s probability distribution and chosen action scored against the same
37 executable state.

38 Executable agent benchmarks verify tool calls against state changes rather than relying only on
39 response judgments [Lu et al., 2024, Yao et al., 2024, Barrès et al., 2025]. Recent voice bench-
40 marks extend this idea to grounded, end-to-end calls: τ -Voice combines full-duplex speech with
41 real-world tools and policies, EVA-Bench evaluates task accuracy and interaction quality across
42 enterprise scenarios, and VAmoS verifies voice-agent behavior against an isolated database [Ray
43 et al., 2026, Bogavelli et al., 2026, Meyer et al., 2026]. These evaluations provide greater interaction
44 and deployment realism. DUAL-AUDIOBENCH concentrates measurement on one mechanism—
45 synchronizing belief after delayed, partially observed change—using matched counterfactuals and
46 explicit probability reports.

47 DUAL-AUDIOBENCH contains 84 scenarios arranged as 42 matched pairs: 14 domains $\times$ three
48 clue-distance buckets $\times$ two causal branches. Each dialogue states a rule and later supplies one
49 fact needed to apply it. In the aligned branch the fact permits completion; in the misaligned branch
50 it creates a recoverable failure. The pair shares fillers, voices, action menus, and post-gap words.
51 Only the early fact changes, causing a different hidden outcome and a different correct final action.
52 Removing that fact makes the paired public histories identical and limits a clue-independent policy
53 to 50% expected post-gap accuracy.

54 After the pre-gap action, a deterministic simulator advances task-specific time, applies any user or
55 external event, and generates the resumed observation from the realized state. Probability reports are
56 collected with the first action, immediately after resumption, and with the final action. This reveals
57 whether new evidence moves probability toward the realized state and separates state-estimation
58 failures from action-selection failures.

59 Our contributions are:

- 60 • **A delayed-transition diagnostic.** Symbolic actions alter a deterministic hidden world;
61 elapsed time and external events produce the evidence and correct action for the realized
62 run.
- 63 • **Matched causal scenarios.** 42 pairs change one early fact while holding the resumed
64 utterance and menu fixed, making clue dependence identifiable rather than inferred from
65 correlation.
- 66 • **Belief and decision measurement.** Three probabilistic checkpoints expose calibration,
67 revision, stale-state mass, and consistency between beliefs, uncertainty flags, and actions.
- 68 • **Controlled audio evaluation.** Nine conditions vary modality, clue distance and removal,
69 external and user-driven change, and spoken delivery with stable menus and inspectable
70 logs.
- 71 • **Auditable evaluation.** Fixed transition rules, plain-language action descriptions, shuffled
72 labels, appropriate chance baselines, and inspectable records make failures reproducible
73 and attributable.

74 **Scope.** DUAL-AUDIOBENCH fixes turn boundaries and replays the visible audio history at each
75 model call. Streaming latency, overlap, barge-in, backchannels, and generated voice quality fall
76 outside its evaluation target; EchoChain and the Full-Duplex-Bench family measure those interac-
77 tion dynamics directly. Here, “long-horizon” refers to causal distance across intervening turns and
78 a simulated time gap, rather than a continuously recorded hour-scale waveform. This scope keeps
79 failures attributable to state inference and belief revision.

80 2 Related Work

81 **Audio and multi-turn understanding.** AIR-Bench, AudioBench, VoiceBench, and WavBench
82 evaluate broad audio comprehension, speech variation, reasoning, and paralinguistics [Yang et al.,
83 2024, Wang et al., 2024, Chen et al., 2024, Li et al., 2026]. BLAB and AudioMarathon study physi-
84 cally long recordings [Ahia et al., 2025, He et al., 2025]. SpokenWOZ supplies dialogue-state anno-
85 tations, while Audio MultiChallenge and MTalk-Bench target multi-turn memory, consistency, natu-
86 ral conversation, and acoustic cues [Si et al., 2023, Gosai et al., 2025, Du et al., 2025]. VoiceAgent-
87 Bench adds multilingual, multi-tool, multi-turn, safety, and adversarial spoken tasks [Jain et al.,

2025]. AudioAgentBench (Audio Arena) uses fixed multi-turn audio to grade tool use, grounding, state tracking, ambiguity handling, and turn taking across customer-facing domains [Arcada Labs, 2026]. These benchmarks establish breadth across audio and dialogue phenomena. DUAL-AUDIOBENCH adds an action-conditioned simulator and scores the model’s probability distribution over the resulting state.

Belief-state representations are well established in task-oriented dialogue, and recent agent work uses explicit uncertainty to separate state estimation from policy learning [Si et al., 2023, Singh et al., 2026]. DUAL-AUDIOBENCH uses the same conceptual separation as an evaluation instrument: the reported distribution is scored at fixed points before and after an executable transition, without training a separate belief model.

Paralinguistic controls. Recent evaluations show that audio access does not guarantee use of vocal affect, and matched-transcript speech delivery can change model behavior [Ayllon et al., 2026, Qian and Li, 2026]. These results make prosody an important control, but they also set a high validation standard: the acoustic manipulation must first be recognizable independently of model behavior. We therefore treat our spoken-delivery conditions as exploratory until the listening audit is complete.

Real-time state revision. Full-Duplex-Bench and its later versions evaluate turn-taking, pacing, corrections, entity tracking, disfluency, and tool use in streaming interaction [Lin et al., 2025b,a, 2026]. MTR-DuplexBench broadens the setting to multi-round instruction following and safety, and FLEXI studies latency, conversational effectiveness, and emergency interruption [Zhang et al., 2025, Ge et al., 2025]. EchoChain most directly overlaps our state-update motivation. It injects a user interruption during assistant speech and diagnoses whether the continuation incorporates the new objective, using a paired half-duplex control [Modi et al., 2026]. EchoChain therefore measures online revision of an answer already in progress. DUAL-AUDIOBENCH measures revision of an explicit world-state distribution after the agent acts and the environment evolves during a gap. The two benchmarks expose different synchronization failures and can be used together.

Grounded voice agents and executable state. ToolSandbox, τ -bench, and τ^2 -bench verify conversational tool use against executable state; the last allows the user and agent to modify a shared environment [Lu et al., 2024, Yao et al., 2024, Barrès et al., 2025]. τ -Voice adds full-duplex interaction, realistic acoustics, policies, and grounded task completion [Ray et al., 2026]. EVA-Bench evaluates complete voice systems through simulated enterprise calls, task and experience metrics, and acoustic perturbations [Bogavelli et al., 2026]. VAmoS uses isolated database instances and trace-level assertions to test containment and policy compliance [Meyer et al., 2026]. These systems more closely approximate deployed calls. DUAL-AUDIOBENCH uses a smaller controlled world so that one changed clue can be tied to a different hidden outcome, gold action, and belief target. By utilizing mechanism-level attribution, it distinguishes failure to infer state from failure to act on a correct inference.

The defining combination in DUAL-AUDIOBENCH is a matched clue intervention that changes an executable terminal state, an implicit post-gap observation, and probability reports scored against the state before and after the transition. This design supports causal claims about clue use and separates belief updating from action selection. The other benchmarks in Table 1 provide complementary evidence about audio robustness, natural interaction, and end-to-end task reliability.

3 Methods

Figure 1 gives the complete evaluation loop. The blue stages are visible to the model and follow half-duplex turn taking: one participant speaks at a time. The gray stage is the hidden simulator. It executes the model’s action, advances time, and uses the resulting state to produce the next spoken observation and answer key.

3.1 Problem formulation

Each scenario has a hidden environment state s_0 . The model chooses a first action a_0 ; during the gap, the user may act, an external event may occur, and task time advances. We summarize this

Table 1: Closest benchmark families and their primary diagnostic target. “State evidence” identifies what supports a success or failure judgment.

Benchmark	Primary target	State evidence	Controlled contrast
Audio MultiChallenge / MTalk [Gosai et al., 2025, Du et al., 2025]	Spoken memory and acoustic cues	Human rubrics / response judgments	Conversation and cue categories
VoiceAgentBench [Jain et al., 2025]	Multilingual spoken tool use and safety	Tool-call correctness	Language, workflow, adversarial query
AudioAgentBench [Arcada Labs, 2026]	Long multi-turn voice-agent behavior	Per-turn category judgments	Domain, TTS vs. human audio
Full-Duplex-Bench / FLEXI [Lin et al., 2025a, Ge et al., 2025]	Streaming timing, correction, interruption	Interaction trace and task scores	Pacing and interruption setting
EchoChain [Modi et al., 2026]	State revision during mid-speech interruption	Instance rubric and human review	Interrupted vs. paired half-duplex
τ -Voice [Ray et al., 2026]	Grounded full-duplex task completion	Executable environment and policy	Text vs. voice; acoustic conditions
EVA-Bench / VAmoS [Bogavelli et al., 2026, Meyer et al., 2026]	End-to-end task quality and containment	Composite metrics / database assertions	Acoustic perturbation / adversarial caller
DUAL- AUDIOBENCH	Delayed belief-state synchronization	Simulator state, belief distribution, and action	Matched clue branch, ablation, and gap controls

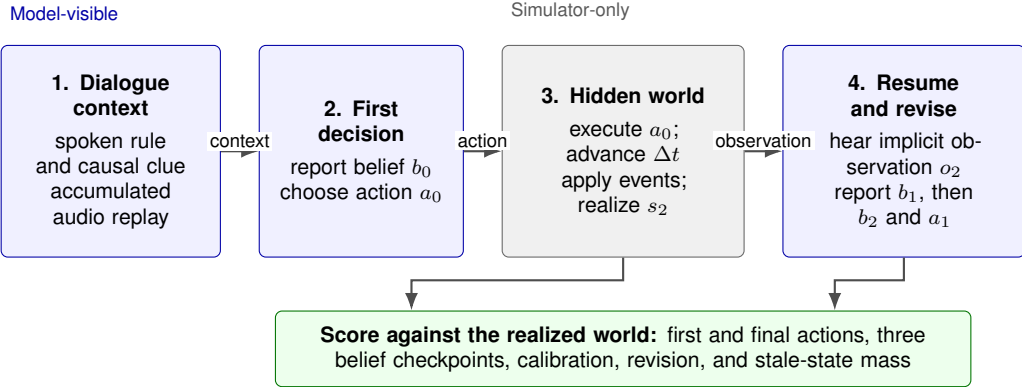


Figure 1: Half-duplex evaluation pipeline. The model hears the accumulated conversation, acts, and later resumes after the hidden environment has changed. Beliefs and actions are evaluated against the same realized state.

138 benchmark-specific relationship as

$$s_2 = T(E(s_0, a_0), u, e, \Delta t), \quad o_2 = O(s_2), \quad a_1^* = G(s_2). \quad (1)$$

139 Here, E executes the model’s action, T advances the world, O generates the resumed observation,
140 and G gives the correct next action. All four functions are deterministic. The model’s first decision
141 can therefore change the state it later encounters and the answer used to score its final decision.

142 At checkpoint $k \in \{0, 1, 2\}$, the model reports a categorical distribution $b_{k,j}$ for every declared
143 hidden-state variable j . Checkpoint 0 is elicited with the pre-gap action and scored against the state
144 immediately before that action executes. Checkpoint 1 follows o_2 and precedes the final menu;
145 checkpoint 2 accompanies the final action. Because belief and action are requested together at
146 checkpoints 0 and 2, these are stated temporal scoring boundaries. The checkpoint-1-to-2 change
147 captures reflection without new user evidence.

148 3.2 Executable scenarios

149 The 84 scenarios cover account access, banking, education, energy, housing, logistics, mobile ser-
150 vice, motor insurance, permits, pharmacy, repair, scheduling, technical support, and travel. In
151 each domain, three clue-distance buckets (1–2, 5–8, and 12–20 turns) are crossed with two causal
152 branches, giving 42 matched pairs. Every task includes an initial state, two action menus, elapsed
153 time, an external event, an optional hidden user action, a short list of possible state values, and fixed
154 rules for updating and scoring the world.

155 Each dialogue states a domain-specific rule before eliciting one policy-relevant fact. In the aligned
156 branch, the fact satisfies the rule, the process completes, and `close_case` is correct. In the
157 misaligned branch, it violates the rule, the process reaches a recoverable terminal state, and a
158 domain-specific recovery action is correct. The model is explicitly asked to report the belief vari-
159 able `causal_alignment`; its values are explained as “the earlier fact satisfies the stated completion
160 rule” and “the earlier fact conflicts with that rule.” The prompt does not reveal the true branch.

161 Within a pair, all public history except the clue turn is identical. Fillers, voices, action text, menu
162 order, and the standard post-gap utterance are held fixed; the latter says that processing ended while
163 withholding its result. The two clues produce different terminal states and different correct final
164 actions. Under clue ablation, the paired public histories become identical. These invariants are
165 checked during generation and by regression tests. Clue distance counts agent and user turns after
166 the clue and before fast-forward, including the first evaluated action and its acknowledgement.

167 Every decision presents five plain-language options labeled A–E. Their internal action names are
168 hidden, and option order changes across repeated runs. Paired conditions always receive the same
169 wording and order. An automated check rejects any scenario where the correct option alone repeats
170 distinctive wording from the clue. Incorrect choices are linked to tentative failure descriptions, such
171 as losing the early clue, retaining an old state, or closing too soon. These descriptions guide analysis
172 but are not treated as human-validated explanations.

173 3.3 Closed-loop interaction protocol

174 One trajectory proceeds as follows:

- 175 1. Initialize s_0 and present the scripted pre-gap user turns in alternating order. For non-
176 decision turns, the evaluated model generates the agent response; a scenario-supplied intent
177 keeps the following user turn coherent but is never replayed as an agent recording.
- 178 2. At the final pre-gap user turn, request a_0 , b_0 , and a Boolean `needs_revalidation` flag.
179 Execute the selected symbolic action against s_0 .
- 180 3. Append a deterministic user acknowledgement. If enabled by the condition, execute a
181 timestamped user action while the agent is absent. Advance Δt and apply the external
182 event to obtain s_2 .
- 183 4. Generate and present o_2 from the realized state. Request b_1 and the revalidation flag without
184 exposing an action menu.
- 185 5. Present the randomized final menu and request b_2 , the final action a_1 , and, for prosodic
186 conditions, a response approach. Execute a_1 and store the full trajectory.

187 In the reflection protocol, the model’s checkpoint-1 distribution is shown back to it at checkpoint 2
188 with an instruction to reconsider it. Consequently, checkpoint-2 action accuracy is a *belief-exposed*
189 measurement. It should not be interpreted as an action-only estimate; an otherwise matched no-
190 belief-exposure ablation is required to quantify any scaffolding introduced by the probe.

191 3.4 Audio-replay conditioning

192 For audio conditions, each user turn and model response is rendered as 16-kHz synthetic speech
193 using eSpeak-NG, with different voices for the two speakers. At each turn, the model hears the
194 visible conversation from the beginning through the current user message. Its own earlier responses
195 are included in that replay. The model answers freely during ordinary dialogue and uses a fixed
196 response format for action choices and beliefs. Speech generation is not evaluated.

197 We evaluate Gemini 2.5 Flash, Gemini 3 Flash Preview, and GPT Audio Mini through OpenRouter,
198 accessed August 10–13, 2026. Requests use a 512-token output limit and $\text{top-}p = 1$; Gemini’s
199 optional reasoning mode is disabled, and other sampling settings use the provider defaults. Two
200 runs per scenario vary menu order and capture ordinary sampling variation.

201 The `transcript_only` condition serializes the same visible history with speaker labels and uses the
202 model’s text interface. The comparison between this condition and `full_audio` therefore measures
203 *audio-replay conditioning versus serialized transcript conditioning*. It should not be described as a
204 pure test of acoustic understanding because the context representation and inference interface change
205 together.

206 Synthetic speech provides deterministic, inexpensive stimuli but is not evidence of robustness to
207 natural voices, accents, channel noise, or disfluency. Core state-tracking claims rely on matched
208 controls and must be replicated on human or higher-fidelity speech before being generalized to
209 deployed voice systems.

210 3.5 Controlled interventions

211 The benchmark defines nine conditions. The three central comparisons use the original audio, re-
212 place the audio with the same written conversation, or remove the early clue while preserving di-
213 alogue length. Additional controls move the clue closer to the decision, suppress the state change
214 during the gap, let the user change the shared state, and vary the delivery of the resumed utterance.
215 Each comparison preserves the scenario, action wording, and option order. The two prosody condi-
216 tions also preserve the words and technical answer.

217 Removing the clue is a validity check. A benchmark that can be solved equally well without its sup-
218 posedly necessary evidence has not shown that models use that evidence. We accordingly compare
219 the original and clue-removed versions of the same scenario and compute uncertainty across the 14
220 domains.

221 3.6 Belief elicitation and scoring

222 At each checkpoint, the model divides its confidence across a short list of possible states whose
223 meanings are provided in the prompt. We measure whether the most likely state is correct, how
224 much confidence the model assigns to the truth, and whether confidence improves after new evidence.
225 Standard measures of probability quality and calibration are reported in the supplementary analysis.

226 Two diagnostics are specific to this benchmark. *Revision gain* is the change in probability assigned
227 to the realized state after the resumed observation. *Stale mass* is the probability that remains on the
228 superseded pre-gap state. We also ask whether the selected action follows from the state the model
229 considered most likely. This separates four outcomes: correct belief and action, correct belief with
230 a poor action, incorrect belief with a lucky action, and a state error that leads to a poor action.

231 3.7 Action, belief, and statistical metrics

232 We report the two action decisions and the three belief checkpoints separately. Strict success requires
233 both actions and every belief checkpoint to be correct; this demanding summary is always shown
234 alongside its components. Each scenario is repeated twice with different option orders and model
235 sampling. Comparisons match the same scenario across conditions, and uncertainty is calculated
236 at the domain level. This prevents several versions of one task family from appearing to provide
237 independent evidence.

238 3.8 Audit and reproducibility protocol

239 Before evaluation, one author reviewed a prespecified 25% sample using randomized, answer-
240 hidden booklets. This construction audit tests whether the public rule and clue determine the causal
241 branch and delayed terminal state; human performance is outside its scope. A second blinded packet
242 used the same 21 scenarios to review one failed trajectory per scenario and mark every supported
243 failure tag. A matched prosody packet remains available for a separate listening check. Inter-rater
244 reliability remains unmeasured.

Table 2: Core results (percent). Every populated cell has $n = 168$. Belief is all-state top-choice accuracy immediately after the gap. Strict success requires both actions and all three belief checkpoints. The complete condition table is in Appendix C.

Model	Ordinary audio replay				Mechanism controls		Explicit user update		Transcript
	First	Belief	Final	Strict	No-change belief	Short-clue belief	Belief	Final	Belief
Gemini 2.5 Flash	69.6	39.3	35.1	1.2	60.7	35.7	72.6	72.0	51.8
Gemini 3 Flash	80.4	51.2	47.0	14.9	78.6	50.6	85.1	79.8	64.3
GPT Audio Mini	65.5	31.0	22.0	0.0	63.1	32.7	67.9	63.1	—

Identical transition inputs produce identical states, and paired conditions receive the same action options in the same order. The released records contain the conversation, model choices, realized states, belief reports, and component scores needed to reproduce every reported result. Full prompt templates, standard metric definitions, and implementation checks are provided with the benchmark materials.

4 Results

4.1 Evaluation coverage

We completed 4,368 trajectories with no API or runtime failures. Gemini 2.5 Flash and Gemini 3 Flash Preview each completed all nine conditions (1,512 trajectories per model); GPT Audio Mini completed the eight audio conditions (1,344 trajectories). Each supported model-condition cell contains two runs of all 84 scenarios. The transcript comparison is limited to the Gemini models because GPT Audio Mini was evaluated only through its audio interface. All results use one fixed scenario release and scoring implementation; development runs are excluded.

The construction audit covered 21 of 84 scenarios across 13 domains, all three clue distances, and 10 aligned and 11 misaligned branches. Using only the answer-hidden booklets, the author recovered both the causal branch and post-gap terminal state in all 21 items, confirming that the delayed outcome is derivable from the public rule and clue. The result establishes internal scenario consistency.

4.2 Models often act correctly before the gap and fail afterward

In audio replay, the accuracy of the first-action ranged from 65.5% to 80.4%, while the accuracy of the final-action ranged from 22.0% to 47.0%. The drop was 34.5 points for Gemini 2.5, 33.4 for Gemini 3, and 43.5 for GPT Audio Mini. This pattern is consistent with the benchmark targeting meaningful post-gap difficulty rather than uniformly difficult action menus. Because the first and final menus ask different questions, this comparison alone does not attribute the entire drop to hidden state change.

Strict success was much lower than action accuracy: 1.2% for Gemini 2.5, 14.9% for Gemini 3, and 0% for GPT Audio Mini in the audio condition. This is not primarily a formatting failure. More than 99% of required belief reports were valid for each model. Models frequently selected plausible actions while assigning their highest confidence to at least one incorrect state. The separate action and belief scores are consequently more informative than the strict composite alone.

4.3 State transition, rather than clue distance, drives the belief gap

When the gap did not change the world, immediate belief accuracy rose by 21.4, 27.4, and 32.1 points for Gemini 2.5, Gemini 3, and GPT Audio Mini, respectively (all $p \leq .030$; Figure 2). Moving the causal clue immediately before the decision did not improve belief accuracy for any model (effects -3.6 , -0.6 , and $+1.8$ points; all $p \geq .72$), nor did it improve the final action. Within the tested turn ranges, the primary difficulty is therefore not explained by simple loss of a distant clue.

Better state estimates did not automatically improve actions. In the no-change control, final-action effects were only $+3.6$, $+1.2$, and -0.6 points, with all intervals crossing zero. This belief-action

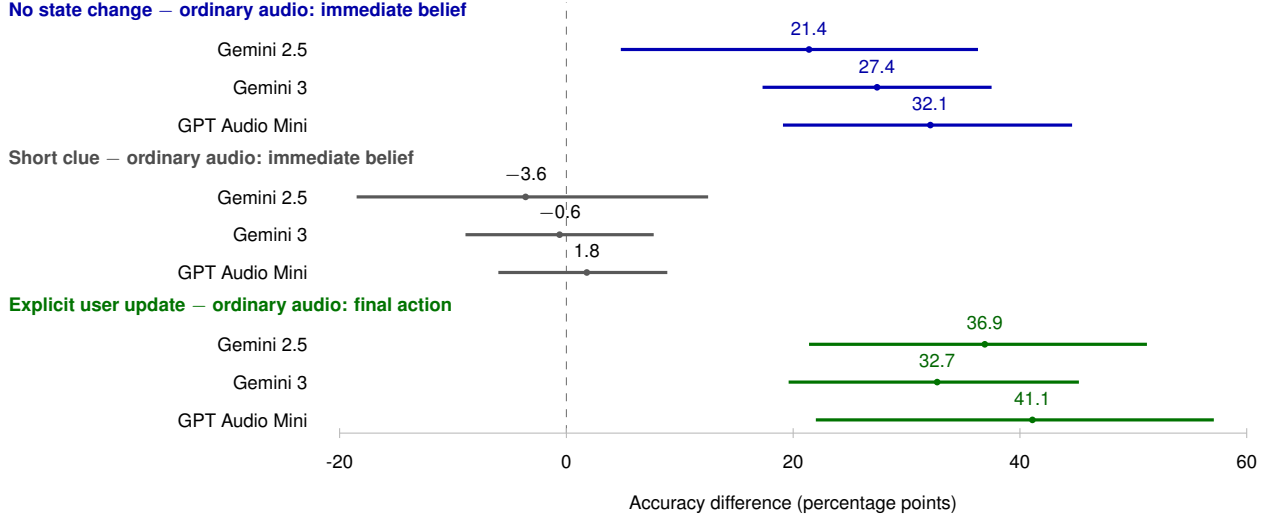


Figure 2: Matched control effects with 95% intervals computed across the 14 domains. Positive values favor the named intervention. Suppressing state change improves belief estimates, shortening clue distance does not, and directly reporting a user-driven update improves the final decision.

Table 3: Paired effect of restoring the causal clue. Effects are percentage points with 95% domain-clustered intervals. Positive values mean that the clue improved performance.

Model	Immediate-belief effect	<i>p</i>	Final-action effect	<i>p</i>
Gemini 2.5 Flash	+9.5 [1.8, 17.3]	.0532	+10.1 [3.6, 17.3]	.0195
Gemini 3 Flash	+19.1 [8.3, 30.4]	.0093	+20.2 [9.5, 30.4]	.0059
GPT Audio Mini	+13.7 [6.0, 21.4]	.0088	-4.8 [-14.3, 5.4]	.4336

separation is precisely what an endpoint-only benchmark would hide. Because the no-change observation truthfully states that processing remains unresolved, the comparison captures the combined burden of a changing world and a less explicit outcome; it does not isolate transition dynamics from observation wording.

4.4 Direct state evidence restores belief and action

In the explicit user-update condition, the user changes the shared state during the gap and the resumed speech directly reports the resulting outcome. Relative to the ordinary, partly implicit transition, immediate belief accuracy improved by 33.3–36.9 points and final-action accuracy improved by 32.7–41.1 points across all three models (all $p \leq .0044$). Thus, the models can revise after a gap when the new state is stated plainly. The contrast changes both the event source and how directly its outcome is described, so it supports an evidence-availability interpretation rather than a claim that user-driven events are intrinsically easier.

4.5 The clue can change belief without changing the decision

Restoring the causal clue improved immediate state estimates for all three models (Table 3). The effect is reliable for Gemini 3 and GPT Audio Mini and borderline under the exact domain-level test for Gemini 2.5. Final-action accuracy, however, improved only for the two Gemini models. GPT Audio Mini assigned more accurate post-gap beliefs when given the clue but did not convert that information into the correct action. This model-level dissociation demonstrates why clue validity and action accuracy should be measured separately.

The branch-pair test is stricter still: a model must answer both counterfactual versions correctly. Gemini 2.5 chose both final actions correctly in 11.9% of pairs with the clue and 1.2% after removal;

304 Gemini 3 achieved 16.7% and 3.6%, respectively. These low absolute rates show that occasionally
305 using the clue is easier than applying the rule consistently across both branches.

306 4.6 Written context improves state estimates more consistently than actions

307 Immediately after the gap, transcript conditioning improved all-state belief accuracy by 12.5 points
308 for Gemini 2.5 and 13.1 points for Gemini 3. The latter paired effect was reliable ($p = .0127$);
309 the former was directionally similar but uncertain ($p = .1104$). Final-action accuracy changed by
310 only 0.6 and 11.3 points, with both intervals crossing zero. The clearest transcript effect therefore
311 concerns the state estimate. This comparison is between accumulated audio replay and a serialized
312 written history.

313 4.7 Spoken-delivery and failure diagnostics

314 The prosody conditions do not yet support a positive grounding claim. Among pairs with valid
315 style responses, accuracy on both matched deliveries was 1.4% for Gemini 2.5, 6.0% for Gemini 3,
316 and 2.2% for GPT Audio Mini, compared with a 6.2% random pair baseline. The directional high-
317 versus-low style contrasts all had intervals crossing zero. Because the one-author listening audit is
318 not complete, these results cannot distinguish model insensitivity from an insufficiently recognizable
319 synthetic-speech manipulation. We report the conditions for completeness but do not use them as
320 evidence for the benchmark’s central claim.

321 Across all conditions, the automatic diagnostics marked an incorrect state belief in 83.5–95.9% of
322 failed trajectories. Early-clue loss appeared in 31.2–52.8%. On the 21 audited failures, the automatic
323 tags achieved 83.8% precision, 55.4% recall, and 0.667 micro F1; exact tag-set agreement was
324 33.3%. The author confirmed all 20 automatically identified state-belief errors, giving that tag 100%
325 precision and recall in the audited sample. Narrower behavioral tags had lower recall, including
326 repeated action (0/6), action-selection failure (1/6), and state-synchronization failure (1/5). We use
327 the state-belief tag as the supported aggregate diagnostic and treat the narrower tags as provisional.

328 5 Discussion and Limitations

329 The replicated control pattern is the central result. All three systems form substantially better state
330 estimates when the world remains unchanged and recover both belief and action when a user-driven
331 outcome is stated directly. Merely shortening clue distance does not help. Together, these results
332 are consistent with a failure to infer a partly observed transition, as opposed to a generic inability
333 to hear the dialogue or remember an earlier fact. The clue ablation adds a second diagnostic: GPT
334 Audio Mini uses the clue in its reported belief but not its action, while the Gemini models show clue
335 effects in both. Endpoint accuracy alone would collapse these distinct behaviors.

336 This target differs from EchoChain’s online revision during interrupted speech and from the end-
337 to-end completion emphasis of τ -Voice, EVA-Bench, and VAmoS. We present a combination of a
338 delayed executable transition, a matched clue intervention, and probability reports scored against
339 the same realized world. The current evidence supports that mechanism-level diagnostic; a potential
340 next step is to mimic natural full-duplex conversation.

341 Several limitations remain. Synthetic speech does not establish robustness to natural voices, accents,
342 noise, or disfluency. The no-change and explicit-user controls necessarily alter what the resumed
343 utterance can truthfully reveal, so they localize the combined inference burden without necessarily
344 isolating a single acoustic variable. Transcript results cover only the Gemini models and change both
345 interface and context representation. The belief-only checkpoint is shown back to the model before
346 its final decision, so final-action accuracy describes belief-elicited behavior. One author audited
347 the construction sample and failure tags; independent human performance and inter-rater agreement
348 remain unmeasured. Finally, prosody should not be fully promoted until undergoing a separate audit
349 process.

6 Conclusion

DUAL-AUDIOBENCH evaluates whether an audio-conditioned agent keeps its view of the world synchronized across a delayed, partly observed transition. Its deterministic environment connects each early clue and action to a realized state, belief target, and required next decision. Across 4,368 trajectories, the models were substantially more successful when no transition occurred or when the changed state was stated directly, while a short clue distance did not resolve the failure. The benchmark therefore exposes a transition-inference gap and reveals belief–action disagreements that task completion alone cannot diagnose.

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438 A Worked causal-pair example

439 Table 4 shows the core intervention using a travel scenario. The spoken rule is the same in both
 440 branches: a departure delay at least as long as the Denver connection window makes the onward
 441 flight impossible unless it is protected. The two conversations differ only in the early answer about
 442 the connection window. After the first action, the simulator applies a 120-minute delay. The resumed

Table 4: A matched causal pair. The early clue changes the realized state and correct action while the post-gap words remain fixed.

Element	Short connection branch	Long connection branch
Early clue	“There are only ninety minutes between the flights in Denver.”	“There are four hours between the flights in Denver.”
Shared event	Departure becomes delayed by 120 minutes.	
Shared resumed words	“The departure is now delayed by 120 minutes, but the notice does not show my connection status. Based on the connection time I mentioned earlier, what should we do next?”	
Realized state	Connection missed	Connection remains viable
Correct action	Protect the later segment and offer a compatible alternative	Conclude that the operation succeeded and close the request

443 words disclose the delay but not the resulting connection status. A model must combine the earlier
 444 clue with the rule to infer the state and choose the final action.

445 The clue-removed condition replaces either early answer with the same uninformative sentence. The
 446 remaining public dialogue, synthetic voices, post-gap words, action descriptions, and randomized
 447 option order are shared within the pair. Because both branches then present the same evidence
 448 but require different answers, a clue-independent policy cannot exceed 50% expected final-action
 449 accuracy on a balanced pair.

450 B Condition definitions

451 The benchmark registers nine conditions, summarized in Table 5. Five test the central state-
 452 synchronization mechanism; the others examine user-driven change and spoken delivery. All nine
 453 were evaluated for both Gemini models. GPT Audio Mini was evaluated on the eight audio condi-
 454 tions.

Table 5: The nine controlled conditions and the factor changed in each one.

Condition	Status	Intervention
Full audio	Three models	Replay the accumulated visible conversation as audio; apply the ordinary gap event.
Transcript only	Two Gemini models	Present the same visible conversation as speaker-labeled text.
Clue removed	Three models	Replace the causal clue with an uninformative answer without changing turn count.
No state change	Three models	Advance through the gap without applying the external state-changing event.
Short distance	Three models	Move the clue pair immediately before the first evaluated decision.
Explicit user update	Three models	Let the user change the shared state while the agent is absent and report the result on return.
Neutral audio	Three models	Render the resumed utterance with neutral delivery.
High prosody	Three models	Render the same words with the pair’s higher-intensity delivery and score response approach.
Low prosody	Three models	Render the same words with the pair’s lower-intensity delivery and score response approach.

455 C Complete condition results

456 Table 6 reports every model–condition cell. The main text emphasizes matched effects because raw
 457 condition differences can combine several changes in evidence and dynamics.

Table 6: Complete condition results (percent). Belief is all-state top-choice accuracy immediately after the gap. Strict success requires both actions and all three belief checkpoints.

Model	Condition	<i>n</i>	First action	Final action	Belief after gap	Strict
Gemini 2.5	Ordinary audio	168	69.6	35.1	39.3	1.2
Gemini 2.5	No state change	168	67.9	38.7	60.7	8.9
Gemini 2.5	Short clue	168	63.7	35.7	35.7	1.8
Gemini 2.5	Clue removed	168	59.5	25.0	29.8	1.2
Gemini 2.5	Transcript	168	77.4	35.7	51.8	10.1
Gemini 2.5	Neutral audio	168	64.9	31.5	32.7	0.6
Gemini 2.5	Explicit user update	168	63.7	72.0	72.6	17.9
Gemini 2.5	High prosody	168	68.5	36.9	37.5	0.6
Gemini 2.5	Low prosody	168	63.7	32.1	35.7	3.6
Gemini 3	Ordinary audio	168	80.4	47.0	51.2	14.9
Gemini 3	No state change	168	75.0	48.2	78.6	21.4
Gemini 3	Short clue	168	66.1	42.3	50.6	11.9
Gemini 3	Clue removed	168	66.7	26.8	32.1	3.6
Gemini 3	Transcript	168	84.5	58.3	64.3	21.4
Gemini 3	Neutral audio	168	78.0	45.2	54.2	10.7
Gemini 3	Explicit user update	168	75.0	79.8	85.1	22.0
Gemini 3	High prosody	168	74.4	44.6	51.2	0.6
Gemini 3	Low prosody	168	72.0	40.5	53.0	2.4
GPT Audio Mini	Ordinary audio	168	65.5	22.0	31.0	0.0
GPT Audio Mini	No state change	168	61.3	21.4	63.1	3.0
GPT Audio Mini	Short clue	168	66.7	20.8	32.7	1.2
GPT Audio Mini	Clue removed	168	55.4	26.8	17.3	0.6
GPT Audio Mini	Neutral audio	168	63.1	19.0	32.1	0.6
GPT Audio Mini	Explicit user update	168	65.5	63.1	67.9	8.9
GPT Audio Mini	High prosody	168	60.7	19.6	30.4	0.0
GPT Audio Mini	Low prosody	168	60.7	20.2	40.5	0.6

D Belief reports in plain language

The model is not asked to invent a hidden state. At each checkpoint, it receives a short, scenario-specific list of possible values and a one-sentence explanation of each value. It assigns probabilities that sum to one. For the travel example, the two reported variables are:

- **Connection status:** *at risk if delayed* means the connection is presently possible but a sufficiently long future delay would break it; *missed* means the present delay is at least the connection window; *protected* means a confirmed rebooking preserves the onward journey; and *viable* means the planned connection remains feasible.
- **Causal alignment:** *misaligned* means the earlier fact conflicts with the completion rule; *aligned* means it satisfies that rule. In the travel pair, these correspond to the ninety-minute and four-hour connection windows. The prompt defines both labels but does not reveal which one is true.

Checkpoint 0 is compared with the state immediately before the first action. Checkpoint 1 is compared with the realized state immediately after the resumed observation. Checkpoint 2 is collected with the final action and checks whether reflection changed the state estimate. This creates a short belief trajectory rather than a single retrospective answer.

E Model request structure

Ordinary conversational turns ask only for a natural-language reply. At an evaluated decision, the request contains five labeled action descriptions, the possible values and explanations for each belief variable, and a request for a probability distribution and revalidation flag. Immediately after the gap, the action menu is withheld so that checkpoint 1 measures state interpretation before the final choices are shown. The final request then supplies the randomized menu and asks for the second post-gap belief report and final action.

Responses are stored in a structured record for automatic scoring, but the paper uses human-readable field names: selected option, confidence over possible states, whether new validation is needed, and

response approach when prosody is evaluated. Missing or malformed fields are counted separately from substantive state errors.

F Metric definitions

Table 7 gives the complete metric set. The main paper emphasizes component accuracies and paired effects because they are easiest to interpret. Probability-quality measures are retained for calibration analysis and reproducibility.

Table 7: Metrics computed from each trajectory.

Metric	Meaning
First-action accuracy	Whether the pre-gap action matches the action required by the initial state.
Final-action accuracy	Whether the post-gap action matches the action required by the realized state.
Belief accuracy	Whether the highest-probability value is correct for every requested state variable at a checkpoint.
Truth probability	The probability assigned to the correct state value; unlike top-choice accuracy, this preserves confidence information.
Negative log-likelihood	Penalizes low probability on the truth, with a particularly large penalty for confident errors.
Brier score	Squared distance between the reported probabilities and the correct one-hot state; lower is better.
Normalized entropy	How diffuse the reported distribution is after accounting for the number of possible values.
Revision gain	Change in probability assigned to the realized state after the resumed observation.
Stale mass	Probability retained on the state that held before the gap but no longer holds.
Belief–action consistency	Whether the selected action is appropriate for the state the model itself considered most likely.
Strict success	Both actions and all three belief checkpoints are correct in the same trajectory.

For matched condition comparisons, each scenario is paired with the same scenario under the intervention. The reported effect is the mean within-domain accuracy difference, and confidence intervals and permutation tests use the 14 domains as the independent units. This avoids treating multiple variants of one domain as unrelated evidence.

G Scenario coverage

Table 8 lists the state variable controlled by the early fact in each domain. Every causal pair has one branch that satisfies the stated completion rule and one that requires a domain-specific recovery action.

Table 8: Domains, outcome variables, and recovery actions in the 84-scenario release.

Domain	Outcome state	Misaligned outcome	Recovery action
Technical support	Firmware status	Stuck	Inspect persistent state
Pharmacy	Claim status	Rejected	Review account configuration
Travel	Connection status	Missed	Protect onward segment
Banking	Dispute status	Returned unmatched	Reconcile card records
Health care	Authorization status	Declined	Obtain qualifying referral
Logistics	Shipment status	Returned to sender	Correct destination record
Energy	Reading status	Flagged incomplete	Register all supply points
Account access	Access status	Still locked	Engage identity provider
Warranty	Claim state	Outside terms	Use retailer route
Housing	Work-order status	Access refused	Reissue access authority
Mobile service	Port status	Rejected	Align ownership record
Education	Enrolment status	Held for review	Request prior-study assessment
Motor insurance	Claim progress	Held for proof	Align policy records
Permits	Permit status	Refused	Supply alternative proof